

AI-Driven Customer Segmentation & Credit Risk Modelling for Financial Decision-Making Report

1. Introduction

In modern financial institutions, understanding customer behaviour and managing credit risk are critical for sustainable decision-making. Organisations must not only identify valuable customers but also assess the likelihood of financial loss from lending activities.

This project aims to address these challenges by applying machine learning techniques to two key areas:

- Customer segmentation based on behavioural data
- Credit risk prediction using financial and loan data

By combining these approaches, the project demonstrates how data-driven insights can improve financial decision-making and risk management.

2. Objectives

The key objectives of this project are:

- To segment customers based on spending and financial behaviour
 - To build a machine learning model to predict loan default risk
 - To identify key factors influencing financial risk
 - To calculate Expected Loss using PD, LGD, and EAD
 - To provide business insights for financial institutions
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3. Datasets

This project uses two main datasets:

3.1 Customer Behaviour Dataset

The credit card dataset contains approximately 9,000 customer records with 18 features, including:

- Balance and credit limit
- Purchase amounts and frequency
- Cash advance usage
- Payment behaviour

This dataset is used for clustering analysis.

3.2 Credit Risk Dataset

The loan portfolio dataset contains 50,000 loan records, including:

- Loan characteristics (sector, type, collateral)
- Financial metrics (leverage, debt-to-equity, interest coverage)
- Risk indicators (PD, LGD, EAD)
- Target variable: defaulted

This dataset is used for supervised machine learning.

4. Methodology

The project is divided into two main parts:

4.1 Customer Segmentation (Unsupervised Learning)

Data Preprocessing

- Removed identifier column (CUST_ID)
- Handled missing values using median imputation
- Standardised features using StandardScaler

Clustering Technique

KMeans clustering was applied to group customers based on behavioural patterns.

Optimal Clusters

The Elbow Method was used to determine the optimal number of clusters.

Cluster Interpretation

Each cluster was analysed based on average feature values.

4.2 Credit Risk Modelling (Supervised Learning)

Data Preparation

- Removed unnecessary columns (IDs and dates)
- Encoded categorical variables using one-hot encoding

Model Selection

A **Random Forest classifier** was used due to:

- Robust performance on structured data
- Ability to capture non-linear relationships
- Reduced risk of overfitting

Model Evaluation

The model was evaluated using:

- Accuracy
- Precision
- Recall
- ROC-AUC
- Confusion Matrix

4.3 Expected Loss Calculation

Expected Loss is calculated using:

$$\text{Expected Loss} = PD \times LGD \times EAD$$

Where:

- **PD** = Probability of Default
- **LGD** = Loss Given Default
- **EAD** = Exposure at Default

This metric represents the expected financial loss for each loan.

5. Results

5.1 Customer Segmentation Results

The clustering analysis identified distinct customer groups:

- **High-value customers:** High spending and high credit usage
- **Low-activity customers:** Minimal transactions and engagement
- **High-risk behaviour group:** Frequent cash advances
- **Moderate users:** Balanced financial behaviour

5.2 Credit Risk Model Results

The model successfully predicted loan default risk.

Key observations:

- Credit score is a strong predictor of default
 - High leverage and debt-to-equity increase risk
 - Financial ratios significantly impact model predictions
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5.3 Feature Importance

The most important features include:

- Credit score
 - Leverage
 - Debt-to-equity ratio
 - PD annual
 - LGD
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5.4 Expected Loss Insights

- Higher PD and LGD values lead to greater expected loss
 - Certain sectors show higher financial risk exposure
 - Expected Loss helps estimate financial impact at portfolio level
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6. Business Insights

The analysis provides valuable insights:

- High cash advance behaviour is linked to higher risk
 - Premium customers can be targeted for loyalty strategies
 - Risky customers can be monitored more closely
 - Sector-based risk analysis can improve portfolio decisions
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7. Business Value

This project can help organisations:

- Improve lending decisions
- Reduce financial losses
- Identify high-risk customers early

- Optimise marketing strategies
 - Enhance portfolio risk management
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8. Ethical Considerations

Machine learning in finance must be used responsibly.

Bias and Fairness

Models may reflect historical bias, leading to unfair decisions.

Privacy

Financial data is sensitive and must be protected under regulations such as GDPR.

Transparency

Model decisions should be explainable to stakeholders.

Accountability

Human oversight is required for critical financial decisions.

9. Limitations

- Synthetic dataset may not fully represent real-world complexity
 - Model performance can be improved with tuning
 - Limited feature engineering
 - No real-time data integration
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10. Future Improvements

- Add SHAP for model explainability
 - Perform hyperparameter tuning
 - Deploy real-time prediction system
 - Integrate additional financial datasets
 - Improve model interpretability
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11. Conclusion

This project demonstrates how machine learning can be applied to both customer segmentation and credit risk modelling.

By combining unsupervised and supervised learning techniques, it provides a comprehensive framework for understanding customer behaviour and assessing financial risk.

The integration of Expected Loss modelling further enhances its practical relevance in real-world financial decision-making.

Overall, the project highlights the importance of combining technical modelling, business understanding, and ethical responsibility in data science.

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